

Response and further development

Supplied v0.2 and research supplement v0.3

Research version 0.3 | 27 September 2026

27 September 2026 — research supplement v0.3

1. What was accepted and preserved

The input v0.2 was treated as the reviewed source, not overwritten. Its full six-job replay passed, including the nested companion exact replay. Forty input files and 130 nested companion files were manifest-verified. The saved equality-search summaries changed only in elapsed-time metadata; the trial records were identical. The new receipt and exact field diff are retained.

The corrections in the supplied review are preserved: Ercan receives attribution for coefficient transport and sign-shift; the six-point extension retains its smaller-cardinality strictness premise; Li–Weston strict descent is cited where used; the restricted negative-type gap does not include the automatic centring zero. The v0.2 seven-point remainder and radius remain the preferred constants for that particular cardinality.

2. New mathematical content

General-cardinality sharp tangent constants. The seven-point active-constraint calculation extends to every binding split with $m \geq 6$ and to the five-point split locally. The constants have six small exceptions and three simple infinite-family formulas. Eighteen exact rational-function dual families prove all higher dimensions; thirty-six rational vectors handle the exceptions. Every active cone direction is integrated into a genuine Ptolemaic curve, so sharpness is geometric rather than only linear-programming sharpness.

Explicit local error and eigendirection bounds. A Taylor and scalar Schur-complement calculation turns the exact cone law into a two-sided bound with an explicit quadratic remainder and radius for each cardinality. The general remainder is intentionally loose. It is not advertised as an improvement of v0.2's sharper seven-point constants.

The metric's own critical exponent. A simple-eigenvalue implicit-function argument gives the first-order change in the supremal negative-type exponent. The two sharp coefficients follow exactly from the cone extremal directions. Strict descent excludes a later return to negative type. This is a fixed-cardinality result with $O(h^2)$ remainder, not an unproved uniform expansion over growing dimensions.

Small cardinalities and quadratic behaviour. Three-point equality is collinearity. Four-point equality is exactly CS(1,3) or CS(2,2), using the external four-point endpoint, strict descent, Ercan's sharp scalar estimate and his transport. An unequal-arm perturbation at the four-point star has gap $p(2p-3)t^{2/8+O(t^4)}$, which rules out a positive linear lower stability constant there. The five-point equality family is not classified by this work, even though the local theorem applies to CS(2,3).

Collision strata. The quotient of a colliding pseudometric identifies the entire kernel at the universal exponent: one zero-sum condition within each collision class. A separate Gaussian-kernel estimate gives an explicit subcritical lower gap for a separated smaller metric. This does not complete a numerical global C_η for the full separated seven-point class.

3. Checks that support the new proof

- Direct symbolic orbit counting verifies the eighteen identities and positive coefficient expansions for all k in each half-line. No fitting or solver call occurs in the proof verifier.
- A separate Fraction reconstruction checks all small duals and original labelled rows, the primal directions and exact feasible curves.

- A separate symbolic calculation reconstructs the four-point second-order eigenvalue coefficient from the six distances.
- All 182 seeded perturbation records and eight 70-digit checks are retained. Numerical evidence is explicitly corroborative.

The certificates are supplied as complete JSON objects and printable tables, not merely as a PASS statement. The numerical discovery history remains labelled and separate.

4. Remaining review obligations

The local tangent and spectral proofs no longer depend on the companion all-size theorem, but their interpretation as a classification of the global equality set still does. Independent specialist review should focus on the orbit-counting dual identity, its universal positivity domains, cone realisability and the critical-exponent identification. Small-cardinality historical priority has not been settled. No formalisation, public release identifier, public licence or authenticated external endorsement has been assigned.